

Assignment 3: Community Market Report

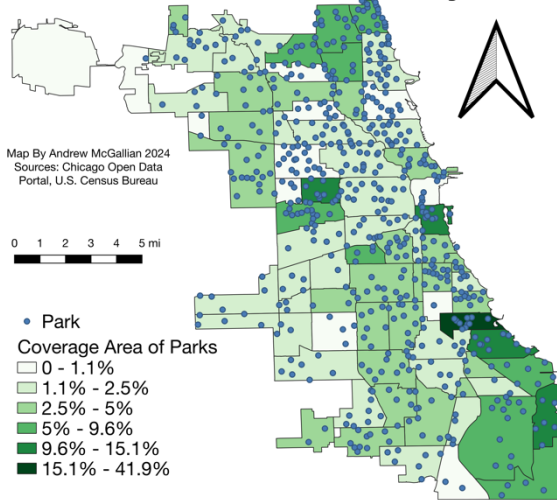
This market analysis focuses on assessing the accessibility and availability of parks within Chicago community areas, with a specific emphasis on identifying areas that are underserved in terms of park resources. The choice of this theme stems from the fundamental role parks play in enhancing community well-being, promoting physical activity, and providing recreational spaces for residents. By utilizing spatial analysis techniques and integrating data from various sources, this analysis aimed to identify areas lacking adequate park infrastructure and highlight the need for targeted interventions. Data for this analysis was obtained from multiple sources. Population and socio-economic data were retrieved from the American Community Survey (1) (ACS) via the tidycensus package in R, while geographic boundaries for community areas (2) and parks (4) were acquired from the Chicago Open Data Portal. Data on households below the poverty line was also sourced from the same portal (3). These datasets facilitated the computation of various metrics, including population density, poverty rates, and park coverage within Chicago Lawn and its neighboring community areas as well as further analysis using GeoDa for guidance on choosing (Appendix f-g).

To evaluate park accessibility, a point-in-polygon (PIP) analysis with R was conducted, focusing on the centroids of park locations within each community area. Additionally, spatial joins with GeoDa and R were performed to aggregate census tract populations within community areas, enabling the calculation of population density and poverty rates. Various maps were generated with QGIS to visualize these metrics, employing Jenks classification for optimal representation, except for park space per person, which required a logarithmic scale due to its wide range of values. Visualizations depict a notable disparity in park accessibility within Chicago Lawn compared to its neighboring community areas. Despite its relatively high

population density, Chicago Lawn exhibits a scarcity of park spaces, ranking significantly lower in park coverage and park space per person (Appendix a-d). Furthermore, the community area falls within the lower quartile for poverty rates, indicating potential socio-economic challenges (Appendix e). The analysis reveals a concerning pattern of inequity in park distribution and socio-economic indicators within Chicago Lawn. The community area appears underserved in terms of recreational amenities, with a significant deficit in park space per person compared to neighboring areas, a “donut” shaped phenomenon. This deficiency not only affects residents' access to leisure activities but also reflects broader disparities in resource allocation and quality of life. Addressing these issues requires targeted interventions to enhance park infrastructure and support community development initiatives. Chicago Lawn presents a clear need for investment in green spaces and socio-economic empowerment programs to promote equitable access to resources and improve residents' well-being.

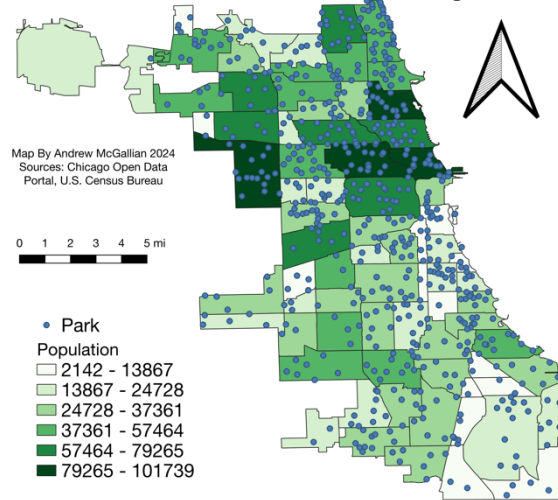
Appendix

Chicago Community Areas and Park Inventory



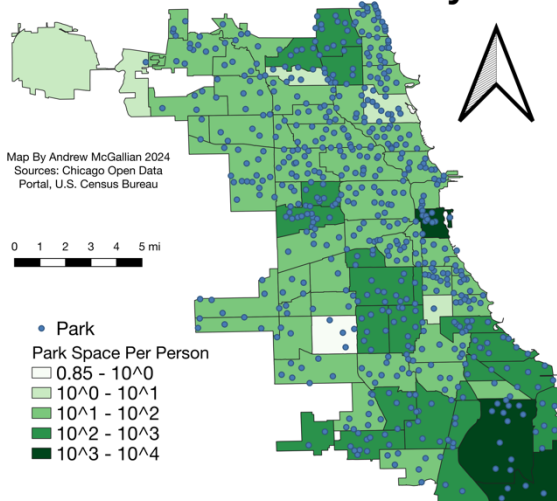
a.

Chicago Community Areas and Park Inventory



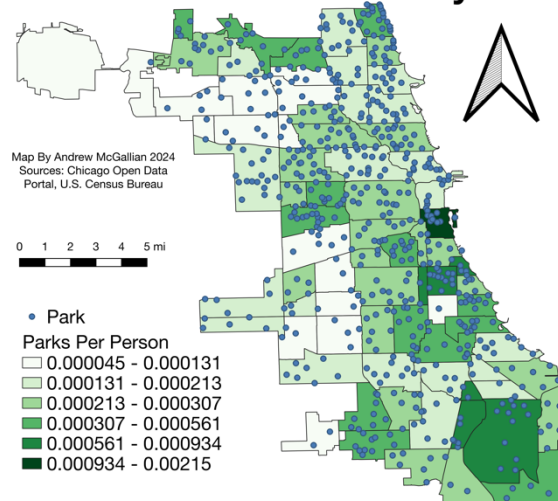
b.

Chicago Community Areas and Park Inventory



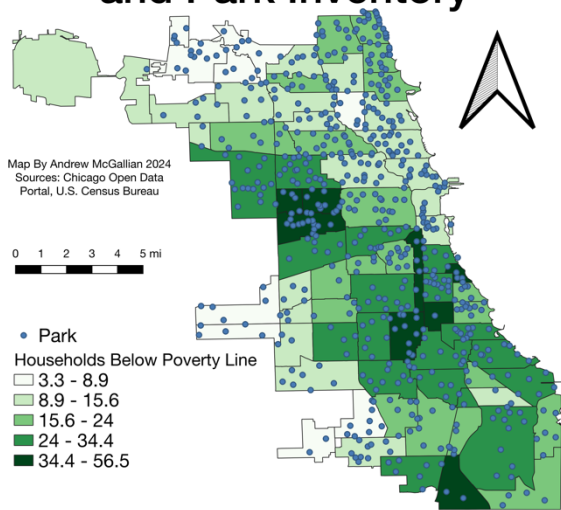
c.

Chicago Community Areas and Park Inventory

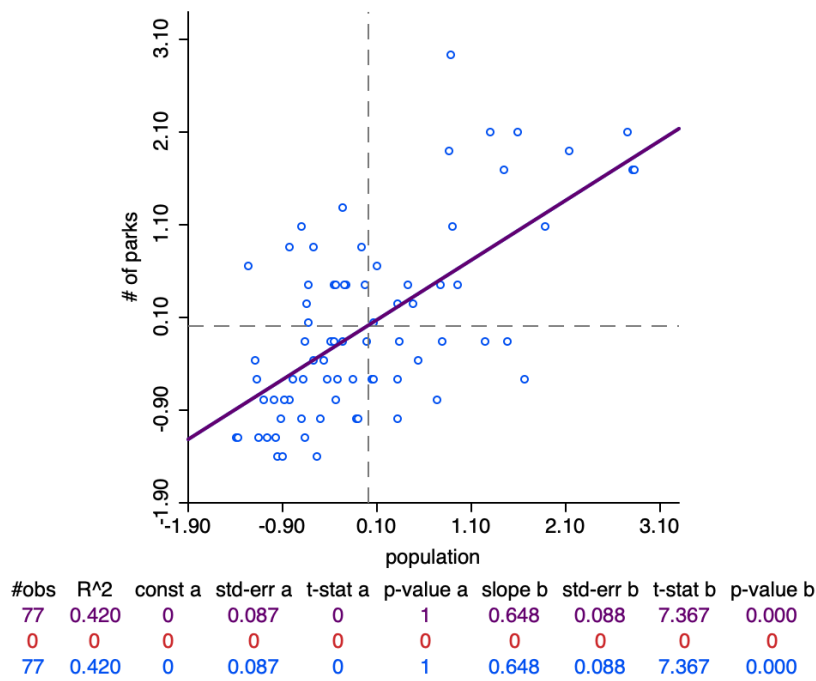


d.

Chicago Community Areas and Park Inventory

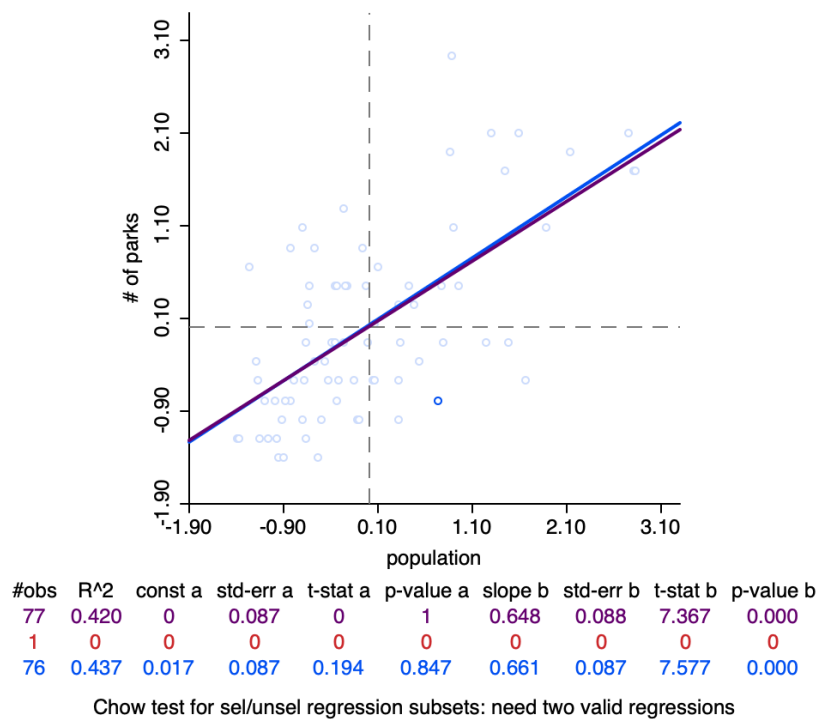


e.



f.

Chow test for sel/unsel regression subsets: need two valid regressions



g. This point is Chicago Lawn, situated in the high population and low number of parks quadrant, prime for investment.

Citations

1. U.S. Census Bureau. (n.d.). tidycensus: Load US Census Boundary and Attribute Data as 'tidyverse' and 'sf'-Ready Data Frames [R package]. Retrieved from <https://cran.r-project.org/web/packages/tidycensus/index.html>
2. "Boundaries - Community Areas (Current): City of Chicago: Data Portal." *Chicago Data Portal*, City of Chicago, 12 Dec. 2023, data.cityofchicago.org/Facilities-Geographic-Boundaries/Boundaries-Community-Areas-current-/cauq-8yn6.
3. Bureau, U.S. Census. "Census Data - Selected Socioeconomic Indicators in Chicago, 2008 – 2012: City of Chicago: Data Portal." *Chicago Data Portal*, City of Chicago, 12 Sept. 2014, data.cityofchicago.org/Health-Human-Services/Census-Data-Selected-socioeconomic-indicators-in-C/kn9c-c2s2/about_data.
4. "Parks - Chicago Park District Park Boundaries (Current): City of Chicago: Data Portal." *Chicago Data Portal*, City of Chicago, 4 Nov. 2016, data.cityofchicago.org/Parks-Recreation/Parks-Chicago-Park-District-Park-Boundaries-current-/ej32-qgdr.